

INDIVIDUAL TECHNICAL REPORT

Environmental problem : High Carbon Emissions

CO900G – Environmental Science and Sustainable Engineering

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1. Abstract

High carbon emissions are one of the major environmental challenges contributing to global warming and climate change. Everyday activities such as electricity consumption, transportation, fuel usage, and waste generation contribute to greenhouse-gas emissions, but many individuals have limited awareness of how their activities affect the environment. To address this issue, this project proposes the development of a responsive web-based Carbon Footprint Calculator designed to help users understand and estimate their personal carbon emissions.

The proposed website provides information about high carbon emissions and identifies individuals, particularly students, households, and general users, as its target audience. The system allows users to enter relevant information about their daily or monthly activities and calculates an estimated carbon footprint using appropriate emission factors. The calculated results are presented in a clear and user-friendly manner, helping users identify the activities that contribute most to their carbon footprint.

The website consists of a minimum of four functional webpages/screens, including an environmental awareness page, a carbon footprint calculator, a results and recommendations page, and a sustainability information page. Interactive features such as the carbon-footprint calculation and personalized emission-reduction recommendations improve user engagement and provide practical environmental guidance. HTML, CSS, and JavaScript are used to develop the responsive and interactive website.

The proposed solution aims to increase environmental awareness, encourage responsible consumption and energy use, and support users in making more sustainable decisions. By providing an accessible digital tool for measuring and understanding carbon emissions, the Carbon Footprint Calculator contributes to climate awareness and supports the broader goal of reducing greenhouse-gas emissions.

2. Introduction

High carbon emissions are a major environmental challenge contributing to global warming and climate change. Carbon dioxide (CO₂) and other greenhouse gases are released through activities such as electricity generation, transportation, fuel consumption, industrial processes, and waste disposal. Reducing these emissions is essential for protecting the environment and supporting sustainable development.

Everyday activities contribute to an individual's carbon footprint, but many people do not clearly understand their environmental impact. Information Technology can help address this issue by providing simple digital tools that convert activity data into estimated carbon emissions. A Carbon Footprint Calculator allows users to understand their emissions and identify activities that contribute most to their environmental impact.

This project focuses on developing a responsive web-based Personalized Carbon Footprint Calculator using HTML, CSS, and JavaScript. Users can enter information related to electricity, transportation, fuel, and waste to calculate their estimated carbon footprint in CO₂e. The system also provides an emission breakdown and personalized recommendations for reducing emissions.

The website consists of four main screens: Home, Calculator, Results and Recommendations, and Sustainability Information. Interactive features allow users to calculate their footprint and receive recommendations based on their results. The target users include students, households, and general individuals. Overall, the project demonstrates how web technologies can be used to increase environmental awareness and encourage more sustainable daily practices.

3. Problem Statement

High carbon emissions from everyday activities such as electricity consumption, transportation, fuel use, and waste generation contribute significantly to climate change. Although many people are aware of environmental issues, they may not know how much carbon their daily activities produce or which activities contribute most to their emissions.

The proposed Personalized Carbon Footprint Calculator addresses this problem by allowing users to estimate their carbon footprint based on their activities. The system identifies major emission sources and provides personalized

recommendations for reducing them. Therefore, the project aims to provide a simple, accessible, and interactive tool that helps individuals understand and reduce their environmental impact.

4. Sustainability Analysis

The proposed Carbon Footprint Calculator is an IT-based solution designed to address high carbon emissions. It supports sustainability by helping users understand how activities such as electricity consumption, transportation, fuel use, and waste contribute to greenhouse-gas emissions. The system estimates users' carbon footprints and provides information that encourages more sustainable behaviour.

4.1 Sustainable Development Goals (SDGs)

The project is mainly aligned with SDG 13 – Climate Action, while also supporting SDG 7 – Affordable and Clean Energy, SDG 11 – Sustainable Cities and Communities, and SDG 12 – Responsible Consumption and Production.

SDG 13 – Climate Action: This is the primary SDG supported by the project. The calculator increases climate awareness by converting users' daily activities into estimated CO₂e emissions. It also provides reduction recommendations, helping users understand how changes in their behaviour can reduce their carbon footprint.

SDG 7 – Affordable and Clean Energy: Electricity consumption is included in the calculator. By showing the emissions associated with energy use, the system encourages users to reduce unnecessary electricity consumption and consider energy-efficient practices and cleaner energy sources.

SDG 11 – Sustainable Cities and Communities: Transportation is an important source of emissions. The calculator allows users to understand the impact of their transportation choices and promotes awareness of lower-carbon options such as public transport, cycling, walking, and shared transportation.

SDG 12 – Responsible Consumption and Production: The system considers electricity, fuel, transportation, and waste consumption. This encourages users to understand the environmental consequences of their consumption and promotes responsible resource use and waste reduction.

Overall, the sustainability process of the website can be represented as:

User Activity → Carbon Calculation → Emission Result → Awareness → Reduction Action

The website therefore contributes indirectly to sustainability by encouraging users to make decisions that may reduce carbon emissions.

4.2 ISO 14001 Analysis

ISO 14001 is an international standard for Environmental Management Systems (EMS). It provides a framework for managing environmental responsibilities and improving environmental performance.

The Carbon Footprint Calculator is not ISO 14001 certified, but its functions relate to several ISO 14001 principles.

First, the system promotes environmental awareness by explaining the relationship between everyday activities and carbon emissions. Second, it supports environmental performance measurement by estimating CO₂e emissions using the formula:

$$\text{Carbon Emission} = \text{Activity Data} \times \text{Emission Factor}$$

The system also helps with the identification of environmental aspects. Electricity consumption, transportation, fuel use, and waste generation are identified as activities that contribute to the user's estimated environmental impact.

The calculator supports continual improvement through its reduction-analysis feature. Users can modify their activity levels and compare their original footprint with a reduced scenario. For example:

$$\text{Potential Reduction} = \text{Original Footprint} - \text{Reduced Footprint}$$

This allows users to understand the possible environmental benefit of changing their behaviour.

The calculated footprint can also act as a baseline for environmental objectives. Users can compare future results with their initial footprint and determine whether their estimated emissions have decreased.

4.3 Life Cycle Assessment (LCA)

Life Cycle Assessment evaluates the environmental impacts of a product, process, or service throughout its life cycle. For this software project, a simplified LCA perspective is used.

During the development stage, computers and other equipment are used for programming, testing, and designing the website, resulting in electricity consumption.

During the deployment stage, servers and hosting infrastructure are required to operate the website. These systems consume electricity and therefore have an indirect environmental impact.

During the usage stage, users access the calculator using computers, tablets, or smartphones. Electricity is consumed by user devices, network infrastructure, and servers.

During the maintenance stage, software updates, bug fixes, security improvements, and content updates require additional computing resources. Efficient coding and optimized website resources can help reduce unnecessary processing and data transfer.

Finally, during the end-of-life stage, computers, smartphones, servers, and other electronic equipment may become electronic waste. Responsible recycling and e-waste management are therefore relevant.

The simplified life cycle can be represented as:

Hardware Production → Development → Hosting → Usage → Maintenance → Hardware End-of-Life

Although the website has an environmental impact through its digital infrastructure, its main sustainability benefit comes from increasing awareness and encouraging users to reduce carbon-intensive activities.

4.4 Environmental Impact Assessment (EIA)

Environmental Impact Assessment (EIA) is generally used for projects involving significant physical, industrial, construction, or infrastructure development.

The proposed Carbon Footprint Calculator is a small-scale software project and does not involve major construction, land development, industrial production, or direct damage to ecosystems. Therefore, a formal EIA is not applicable.

However, environmental impacts related to the IT system have been considered. These include electricity consumption by computers and servers, network energy use, data-centre resource consumption, electronic-device manufacturing, and electronic waste.

Considering these impacts ensures that the project recognizes the environmental footprint of digital technology while focusing on its intended positive contribution to carbon-emission awareness and reduction.

4.5 Overall Sustainability Assessment

The Carbon Footprint Calculator demonstrates strong alignment with sustainability objectives.

Area	Project Contribution
SDG 13	Climate awareness and carbon-emission reduction
SDG 7	Energy-use awareness and efficiency
SDG 11	Awareness of sustainable transportation
SDG 12	Responsible consumption and waste reduction
ISO 14001	Environmental awareness, measurement, environmental aspects, and continual improvement

Area	Project Contribution
LCA	Considers environmental impacts throughout the software life cycle
EIA	Formal EIA is not applicable to this small-scale software project

In conclusion, the proposed Carbon Footprint Calculator provides a practical IT-based approach to environmental sustainability. It supports SDG 13 as its primary sustainability goal and contributes to SDGs 7, 11, and 12. Although the website itself has a small environmental footprint through energy and digital infrastructure use, its main purpose is to encourage users to understand and reduce their carbon emissions. The combination of carbon measurement, awareness, and reduction analysis makes the system a relevant technology-based solution for addressing high carbon emissions.

5. Proposed Engineering/IT Solution

5.1 Personalized Carbon Footprint Calculator

The proposed Engineering/IT solution for addressing the problem of high carbon emissions is a Personalized Carbon Footprint Calculator. It is a responsive web-based application designed to help individuals understand and estimate the greenhouse-gas emissions produced by their everyday activities. The system focuses on providing users with a simple way to calculate their estimated carbon footprint and receive personalized recommendations for reducing emissions.

High carbon emissions are associated with several common activities, including electricity consumption, transportation, fuel usage, and waste generation. Many individuals are aware that these activities affect the environment but may not understand the amount of emissions associated with their personal behaviour. The proposed system addresses this problem by converting user-provided activity information into an estimated carbon footprint.

The calculator uses a simple calculation model:

$$\text{Carbon Emissions} = \text{Activity Data} \times \text{Emission Factor}$$

The system collects relevant information from the user, such as electricity consumption, distance travelled, type of transportation, fuel consumption, and waste generation. This information is processed using predefined emission factors to calculate estimated carbon dioxide equivalent (CO₂e) emissions.

The total carbon footprint can be represented as:

$$\text{Total Carbon Footprint} = \text{Electricity Emissions} + \text{Transportation Emissions} + \text{Fuel Emissions} + \text{Waste Emissions}$$

The result is displayed in an understandable format so that users can identify which activities contribute most to their overall carbon footprint.

5.2 Personalized Reduction Suggestions

A key feature that makes the proposed solution different from a basic carbon calculator is its personalized reduction recommendation system. After calculating the user's footprint, the system identifies the categories with relatively high emissions and provides relevant suggestions.

For example, if transportation represents the largest part of a user's footprint, the system may provide recommendations such as using public transportation, walking, cycling, carpooling, or reducing unnecessary vehicle trips. If electricity is the largest source, the system can recommend reducing unnecessary electricity consumption and using energy-efficient practices.

The recommendations are therefore connected to the user's calculated results rather than providing the same general advice to every user.

The basic process is:

User Data → Carbon Calculation → Emission Analysis → Identify High-Emission Category → Personalized Recommendation

This approach makes the system more useful because users can immediately connect the recommendations to their own behaviour.

5.3 Main System Features

The proposed Carbon Footprint Calculator contains several functional features.

1. User Input:

Users enter information about their daily or monthly activities. The main categories include electricity, transportation, fuel, and waste.

2. Carbon Calculation:

JavaScript processes the entered information and applies the relevant emission factors to estimate the user's carbon footprint.

3. Emission Breakdown:

The system displays the contribution of each category separately. This allows users to identify their major sources of emissions.

4. Personalized Recommendations:

The system provides reduction suggestions based on the categories that contribute most to the user's footprint.

5. Reduction Comparison:

Users can modify their activity levels and observe how reducing certain activities may affect their estimated footprint.

6. Responsive Interface:

The website is designed to work on desktops, laptops, tablets, and smartphones, allowing users to access the calculator from different devices.

5.4 Website Structure

The proposed solution can be implemented through four main webpages or screens.

Page 1 – Home Page:

The home page introduces the problem of high carbon emissions and explains why measuring personal carbon footprints is important. It also provides a clear entry point to the calculator.

Page 2 – Carbon Footprint Calculator:

Users enter their activity information, including electricity consumption, transportation, fuel, and waste. A calculation button processes the information and produces the estimated footprint.

Page 3 – Results and Recommendations:

The results page displays the total estimated carbon footprint and provides a breakdown of emissions by category. Personalized recommendations are displayed based on the user's highest-emission activities.

Page 4 – Sustainability/Awareness Page:

This page provides information about climate change, sustainable practices, relevant Sustainable Development Goals, and simple methods of reducing carbon emissions.

5.5 Interactive Functionality

The proposed website includes interactive features that improve user engagement.

The first major interactive feature is the carbon footprint calculation. When users enter their activity data and submit the form, JavaScript calculates and displays the estimated emissions without requiring the page to be reloaded.

The second feature is personalized recommendations. The system analyses the calculated categories and displays suggestions according to the user's highest-emission areas.

A further interactive feature is what-if reduction analysis. Users can reduce selected activity values and recalculate their footprint to see the potential difference.

For example:

Original Footprint = 100 kg CO₂e

After reducing transportation and electricity consumption:

Reduced Footprint = 80 kg CO₂e

Therefore:

Potential Reduction = 100 – 80 = 20 kg CO₂e

Reduction Percentage = 20%

This feature helps users understand that small changes in daily behaviour can potentially contribute to lower emissions.

5.6 Technologies Used

The proposed system can be developed using standard web technologies:

- HTML5 – Used to create the structure and content of the webpages.
- CSS3 – Used for styling, layout, responsive design, and visual presentation.
- JavaScript – Used for carbon calculations, form validation, interactive results, recommendation generation, and what-if analysis.

These technologies are suitable because the calculator mainly requires user input, mathematical calculations, data processing, and dynamic presentation of results. The system can operate as a client-side web application without requiring a complex backend.

5.7 Expected Output

The main output of the system is the user's estimated carbon footprint, expressed in kg or tonnes of CO₂e for the selected period. The system also presents an emission breakdown and personalized recommendations.

For example, a result may show:

Category	Estimated Emissions
Electricity	35 kg CO ₂ e
Transportation	45 kg CO ₂ e
Fuel	15 kg CO ₂ e
Waste	5 kg CO ₂ e
Total	100 kg CO ₂ e

The user can then see that transportation is the largest contributor and receive recommendations focused on reducing transportation emissions.

5.8 Environmental Contribution

The main environmental contribution of the Personalized Carbon Footprint Calculator is awareness and behaviour change. The system does not directly remove carbon dioxide from the atmosphere. Instead, it provides users with information that can help them make more environmentally responsible decisions.

By identifying high-emission activities and providing personalized reduction suggestions, the system can encourage lower electricity consumption, more sustainable transportation choices, reduced fuel use, and better waste practices.

The solution therefore connects technology with environmental action:

Measure → Understand → Identify → Reduce → Improve

This makes the Personalized Carbon Footprint Calculator a practical IT solution for addressing high carbon emissions at an individual level.

5.9 Summary

The Personalized Carbon Footprint Calculator provides a simple, interactive, and accessible method for individuals to estimate and understand their carbon emissions. Unlike a basic information website, the proposed system uses user-specific activity data to calculate emissions and generate relevant reduction suggestions.

The combination of carbon calculation, emission breakdown, personalized recommendations, and what-if analysis makes the system useful for promoting sustainable behaviour. Developed using HTML, CSS, and JavaScript, the solution is technically feasible and suitable for a responsive educational website. Its main objective is to transform carbon-emission information into practical actions that can contribute to reducing individual environmental impact.

6. Feasibility Analysis

The proposed Personalized Carbon Footprint Calculator is a web-based IT solution designed to calculate users' estimated carbon emissions and provide personalized recommendations for reducing their environmental impact. The feasibility analysis considers whether the system can be developed successfully within the available technical, financial, time, and resource constraints.

6.1 Technical Feasibility

The system is technically feasible because it can be developed using HTML5, CSS3, and JavaScript. HTML provides the webpage structure, CSS supports responsive design, and JavaScript performs calculations, input validation, emission analysis, and personalized recommendations. The basic calculation is:

Carbon Emissions = Activity Data × Emission Factor

The system does not require advanced hardware, complex databases, or specialised software and can operate through a standard web browser.

6.2 Economic Feasibility

The project has a low development cost because HTML, CSS, JavaScript, and common development tools are freely available. It can be developed using existing computers and free software. If deployed publicly, costs may include hosting, domain registration, and internet access; however, these costs are relatively low. Therefore, the solution is suitable for a student project with limited financial resources.

6.3 Operational Feasibility

The system is designed to be simple and accessible to general users. Users enter information related to electricity, transportation, fuel, and waste, after which the system automatically calculates their estimated carbon footprint and provides recommendations. Its responsive design allows access through smartphones, tablets, laptops, and desktops. The four-page structure consists of a Home Page, Calculator Page, Results Page, and Sustainability Page, making the system easy to navigate and understand.

6.4 Environmental and Social Feasibility

The solution has a positive environmental purpose because it promotes awareness of personal carbon emissions and encourages lower-carbon practices such as reducing electricity consumption, using public transport, walking, cycling, reducing fuel use, and minimizing waste.

Socially, the calculator can benefit students, households, and general users by presenting environmental information in a simple and personalized manner. It can also serve as an educational tool in schools and universities.

6.5 Legal and Ethical Feasibility

The system can be implemented responsibly because basic carbon calculations do not require sensitive personal information. User data should be handled appropriately, and calculated results should clearly be presented as estimates based on selected emission factors. External information and resources should also be properly acknowledged.

6.6 Overall Feasibility

Overall, the Personalized Carbon Footprint Calculator is technically, economically, operationally, environmentally, and socially feasible. Its simple technologies, low cost, manageable development requirements, and environmental purpose make it suitable for implementation as an academic IT project. The system can provide useful carbon-awareness information while remaining practical within the available resources and project timeframe.

7. Justification of the Proposed Solution

The Personalized Carbon Footprint Calculator was selected because high carbon emissions are strongly influenced by everyday activities such as electricity use, transportation, fuel consumption, and waste generation. Many individuals may be aware of climate change but may not understand how their personal activities contribute to carbon emissions. The proposed system addresses this problem by allowing users to calculate their estimated carbon footprint in a simple and accessible way.

A key advantage of the solution is its personalized recommendations. Instead of providing only a total emission value, the system identifies the user's major emission categories and suggests practical actions for reducing them. This makes the information more relevant and useful to each individual.

The solution is also appropriate for an academic IT project because it can be developed using HTML, CSS, and JavaScript without requiring expensive hardware or complex infrastructure. Its interactive features, responsive design, and simple user interface make it suitable for students, households, and general users.

Therefore, the proposed solution is justified because it combines carbon measurement, environmental awareness, and practical reduction guidance in one accessible web-based system. It provides a realistic and manageable technological approach to supporting individual action toward reducing carbon emissions.

8. Expected Environmental Benefits

The proposed Personalized Carbon Footprint Calculator is expected to provide several environmental benefits by helping users understand and reduce the carbon emissions associated with their daily activities. The main environmental benefit is increased awareness of personal carbon footprints. By entering information about electricity consumption, transportation, fuel usage, and waste generation, users can identify which activities contribute most to their emissions.

The system can encourage users to adopt low-carbon habits by providing personalized recommendations based on their highest-emission category. For example, users with high transportation emissions may be encouraged to use public transport, walk, cycle, or share vehicles. Users with high electricity-related emissions can be encouraged to reduce unnecessary electricity consumption and use energy-efficient appliances. These behavioural changes can contribute to lower greenhouse gas emissions over time.

Another important benefit is the comparison of possible reduction scenarios. Users can examine how changes in their daily activities may affect their overall carbon footprint. This makes environmental action more measurable and motivates users to choose practical reduction strategies. Although the calculator does not directly remove carbon emissions, it supports informed decision-making that can lead to emission reductions.

The solution also supports sustainable resource use by increasing awareness of electricity, fuel, transportation, and waste consumption. It can be particularly useful in educational environments, where students can learn about the relationship between everyday activities and environmental impacts.

Overall, the proposed system provides an indirect but meaningful environmental benefit by connecting measurement, awareness, and action. If users regularly apply the recommendations provided by the calculator, the combined effect of individual behavioural changes can contribute to reduced carbon emissions and support broader climate-change mitigation efforts.

9. Conclusion

High carbon emissions are a significant environmental challenge that requires both technological solutions and changes in individual behaviour. This project proposed a Personalized Carbon Footprint Calculator as a practical web-based solution for increasing awareness of personal carbon emissions and encouraging sustainable lifestyle choices.

The proposed system allows users to enter information related to electricity consumption, transportation, fuel usage, and waste, and calculates their estimated carbon footprint. It further provides personalized recommendations based on the user's major emission categories, making the system more useful than a basic calculation tool. The interactive and responsive design also makes the solution accessible to students, households, and general users.

The feasibility analysis demonstrated that the system can be developed using commonly available technologies such as HTML, CSS, and JavaScript with relatively low cost and technical requirements. The solution also supports sustainability objectives by promoting energy efficiency, responsible transportation, reduced fuel consumption, and waste reduction.

Overall, the Personalized Carbon Footprint Calculator provides a simple, affordable, and practical IT approach to addressing high carbon emissions. By connecting measurement, awareness, and action, the system can encourage users to make more sustainable decisions and contribute to long-term carbon-emission reduction.

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